# Equivalent fractions

**1** Select the equivalent fractions from the following lists:

**a** $\frac{4}{40}, \frac{1}{10}, \frac{2}{80}, \frac{8}{80}, \frac{8}{20}$

**b** $\frac{20}{30}, \frac{7}{3}, \frac{8}{12}, \frac{2}{6}, \frac{2}{3}$

**c** $\frac{1}{4}, \frac{16}{64}, \frac{4}{8}, \frac{128}{256}, \frac{32}{64}$

**d** $\frac{9}{8}, \frac{27}{24}, \frac{108}{96}, \frac{8}{9}, \frac{54}{24}$

**e** $\frac{15}{20}, \frac{6}{8}, \frac{10}{15}, \frac{9}{12}, \frac{2}{3}$

**f** $\frac{6}{9}, \frac{3}{4}, \frac{10}{15}, \frac{17}{24}, \frac{4}{6}$

**g** $\frac{4}{18}, \frac{14}{63}, \frac{2}{16}, \frac{2}{9}, \frac{10}{36}$

**h** $\frac{17}{5}, \frac{8}{20}, \frac{2}{5}, \frac{4}{14}, \frac{16}{40}$

**2** Complete the following equivalent fraction statements:

**a** $\frac{1}{2} = \frac{\square}{4}$

**b** $\frac{1}{5} = \frac{\square}{75}$

**c** $\frac{3}{8} = \frac{\square}{40}$

**d** $\frac{9}{5} = \frac{\square}{40}$

**e** $\frac{19}{10} = \frac{\square}{90}$

**f** $\frac{3}{8} = \frac{\square}{24}$

**g** $\frac{5}{16} = \frac{20}{\square}$

**h** $\frac{7}{12} = \frac{35}{\square}$

**i** $\frac{4}{9} = \frac{40}{\square}$

**j** $\frac{4}{3} = \frac{48}{\square}$

**k** $\frac{5}{21} = \frac{45}{\square}$

**l** $\frac{18}{11} = \frac{72}{\square}$

**3** Complete the table below:

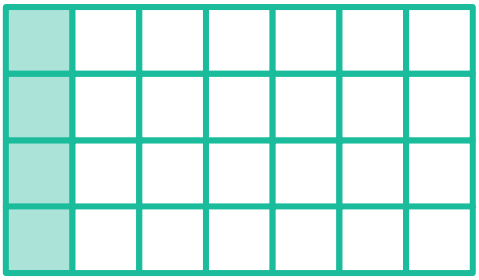
| Mixed number | Improper fraction | Equivalent fraction |
|---|---|---|
| $2\frac{1}{6}$ | | |
| $3\frac{4}{5}$ | | |
| $4\frac{2}{7}$ | | |
| $1\frac{2}{5}$ | | |
| $6\frac{1}{8}$ | | |
| $2\frac{3}{8}$ | | |
| $3\frac{3}{7}$ | | |
| $4\frac{2}{9}$ | | |
| $5\frac{1}{4}$ | | |

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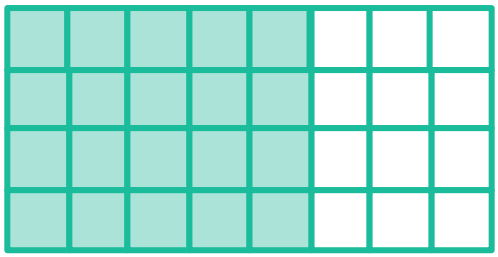
## Simplify fractions

**4** Use the diagrams provided to simplify the following fractions:

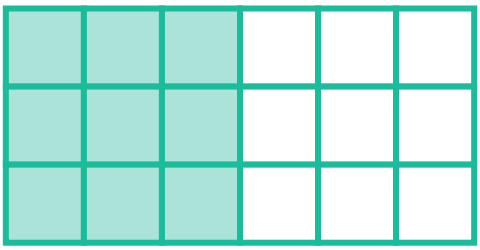
**a** $\frac{4}{28}$



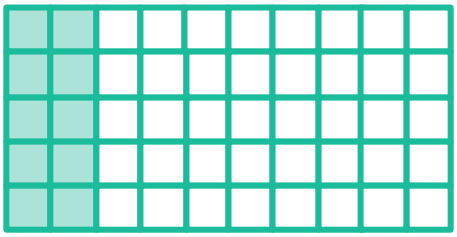
**b** $\frac{20}{32}$



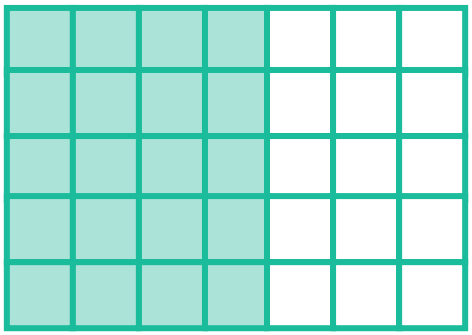
**c** $\frac{9}{18}$



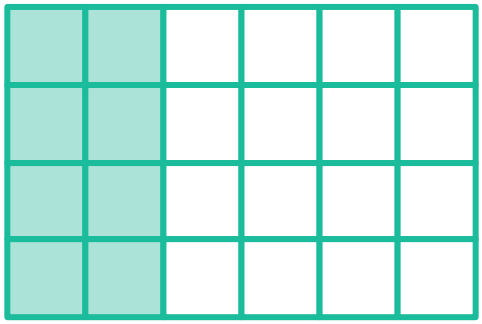
**d** $\frac{10}{50}$



**e** $\frac{20}{35}$



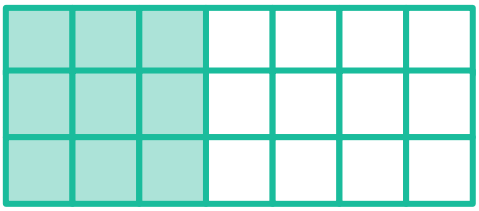
**f** $\frac{8}{24}$

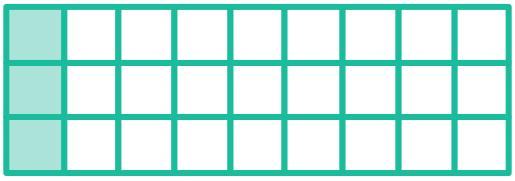


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g $\frac{9}{21}$

h $\frac{3}{27}$





**5** Simplify the following fractions:

a $\frac{3}{9}$ b $\frac{20}{60}$ c $\frac{6}{20}$ d $\frac{27}{45}$

e $\frac{63}{28}$ f $\frac{40}{30}$ g $\frac{26}{50}$ h $\frac{24}{66}$

i $\frac{32}{80}$ j $\frac{64}{92}$ k $\frac{80}{100}$ l $\frac{60}{200}$

m $\frac{2}{10}$ n $\frac{8}{64}$ o $\frac{21}{231}$ p $\frac{150}{270}$

## Compare fractions

**6** For the each of the following pairs of fractions, state which fraction is bigger:

a $\frac{1}{3}, \frac{3}{15}$ b $\frac{4}{28}, \frac{6}{14}$ c $\frac{31}{8}, \frac{24}{7}$ d $\frac{3}{8}, \frac{1}{2}$

e $\frac{14}{5}, \frac{11}{4}$ f $\frac{5}{2}, \frac{49}{20}$ g $1\frac{6}{7}, \frac{25}{14}$ h $3\frac{4}{9}, \frac{7}{2}$

**7** For the each of the following pairs of fractions, state which fraction is smaller:

a $\frac{2}{9}, \frac{7}{8}$ b $\frac{7}{5}, \frac{8}{10}$ c $\frac{4}{15}, \frac{2}{5}$ d $\frac{4}{7}, \frac{4}{8}$

e $\frac{63}{15}, \frac{82}{20}$ f $\frac{21}{6}, \frac{18}{5}$ g $\frac{27}{45}, \frac{5}{8}$ h $2\frac{5}{9}, 2\frac{7}{11}$

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**8** For each of the following groups of fractions:

i Find the lowest common denominator of the 3 fractions.

ii Rewrite all 3 fractions with this lowest common denominator.

iii List the 3 fractions in ascending order.

**a** $\frac{2}{5}, \frac{9}{55}, \frac{7}{11}$ **b** $\frac{7}{13}, \frac{8}{65}, \frac{3}{5}$ **c** $\frac{5}{7}, \frac{9}{77}, \frac{8}{11}$ **d** $\frac{5}{17}, \frac{7}{68}, \frac{3}{4}$

**9** Arrange the following groups of fractions from smallest to largest:

**a** $\frac{11}{20}, \frac{4}{15}, \frac{2}{5}$ **b** $\frac{2}{3}, \frac{3}{7}, \frac{4}{5}$ **c** $\frac{21}{9}, \frac{6}{4}, \frac{2}{7}$ **d** $4\frac{1}{2}, \frac{31}{4}, \frac{15}{2}$

**e** $\frac{39}{4}, 4\frac{5}{10}, \frac{19}{2}$ **f** $1\frac{6}{7}, \frac{33}{14}, \frac{16}{7}$ **g** $2\frac{2}{5}, \frac{33}{10}, \frac{16}{5}$ **h** $\frac{17}{3}, \frac{35}{6}, 3\frac{2}{3}$

**10** For each of the following groups of fractions:

i Find the lowest common denominator of the 3 fractions.

ii Rewrite all 3 fractions with this lowest common denominator.

iii List the 3 fractions in descending order.

**a** $\frac{1}{4}, \frac{3}{8}, \frac{5}{32}$ **b** $\frac{2}{6}, \frac{4}{9}, \frac{1}{54}$ **c** $\frac{2}{5}, \frac{6}{7}, \frac{8}{35}$ **d** $2\frac{2}{5}, \frac{37}{10}, \frac{18}{5}$

**11** Arrange the following groups of fractions from largest to smallest:

**a** $\frac{1}{2}, \frac{2}{3}, \frac{5}{8}$ **b** $\frac{3}{8}, \frac{9}{16}, \frac{13}{24}$ **c** $2\frac{7}{10}, \frac{13}{5}, \frac{7}{2}$ **d** $2\frac{1}{7}, \frac{33}{14}, \frac{16}{7}$

**e** $6\frac{1}{2}, \frac{35}{4}, \frac{17}{2}$ **f** $\frac{33}{10}, 1\frac{4}{5}, \frac{16}{5}$ **g** $\frac{17}{4}, 4\frac{3}{4}, \frac{39}{8}$ **h** $\frac{21}{2}, \frac{37}{4}, 9\frac{1}{2}$

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